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ROLL NO.: A-18

BATCH:-A1

ASSINGMENT:-7

AIM: Implement and analyze various probability distributions in Python, including Binomial, Poisson, and Normal distributions.

1. Difference between Binomial, Poisson, and Normal distributions

- Binomial Distribution is a discrete distribution used when there is a fixed number of trials and each trial has only two outcomes (success or failure) with constant probability.
- Poisson Distribution is a discrete distribution used to model the number of events occurring in a fixed interval of time or space when events happen independently.
- Normal Distribution is a continuous distribution that is symmetric and bell-shaped, defined by mean and standard deviation.

In short:

Binomial → fixed trials Poisson → event counts Normal → continuous data

2. Real-world significance of Poisson distribution

Poisson distribution is used to model real-life situations where we count how many times an event occurs in a given interval.

Examples:

Number of calls received in a call center Number of accidents in a day Number of website visitors per minute

It is important because it helps in predicting and managing systems like traffic, networks, and queues.

3. Central Limit Theorem (CLT)

The Central Limit Theorem states that:

When the sample size is large, the distribution of the sample mean tends to become normal, regardless of the original distribution.

This means even if the data is not normally distributed, the average of samples will follow a normal distribution.

4. Why Normal distribution is widely used

Normal distribution is widely used because:

Many natural and real-world data follow this pattern It is mathematically simple to work with It is used in many statistical methods and machine learning models

Examples:

Heights of people Exam scores Measurement errors **5. What is skewness and which distributions exhibit it**

Skewness refers to the asymmetry of a distribution.

Positive skew → tail on right side Negative skew → tail on left side Zero skew → symmetric distribution

Distribution behavior:

Binomial → can be skewed or symmetric Poisson → usually positively skewed Normal → no skew (perfectly symmetric)

```
import pandas as pd
```

```
df = pd.read_csv("/content/train (1).csv")
```

```
print(df.head())
```

```
print(df.info())
```

```

  PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

```

```

      Name      Sex  Age  SibSp  \
0  Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2  Heikkinen, Miss. Laina    female  26.0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0      1
4  Allen, Mr. William Henry    male  35.0      0

```

```

   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN      S
1      0    PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282   7.9250   NaN      S
3      0    113803  53.1000  C123      S
4      0    373450   8.0500   NaN      S

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
```

```

#   Column      Non-Null Count  Dtype
---  -
0  PassengerId  891 non-null      int64
1  Survived     891 non-null      int64
2  Pclass       891 non-null      int64
3  Name         891 non-null      object
4  Sex          891 non-null      object

```

```
5   Age          714 non-null   float64
6   SibSp        891 non-null   int64
7   Parch        891 non-null   int64
8   Ticket       891 non-null   object
9   Fare         891 non-null   float64
10  Cabin        204 non-null   object
11  Embarked     889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

```
data = df.select_dtypes(include=['number']).iloc[:, 0]

print("Selected column:", data.name)
```

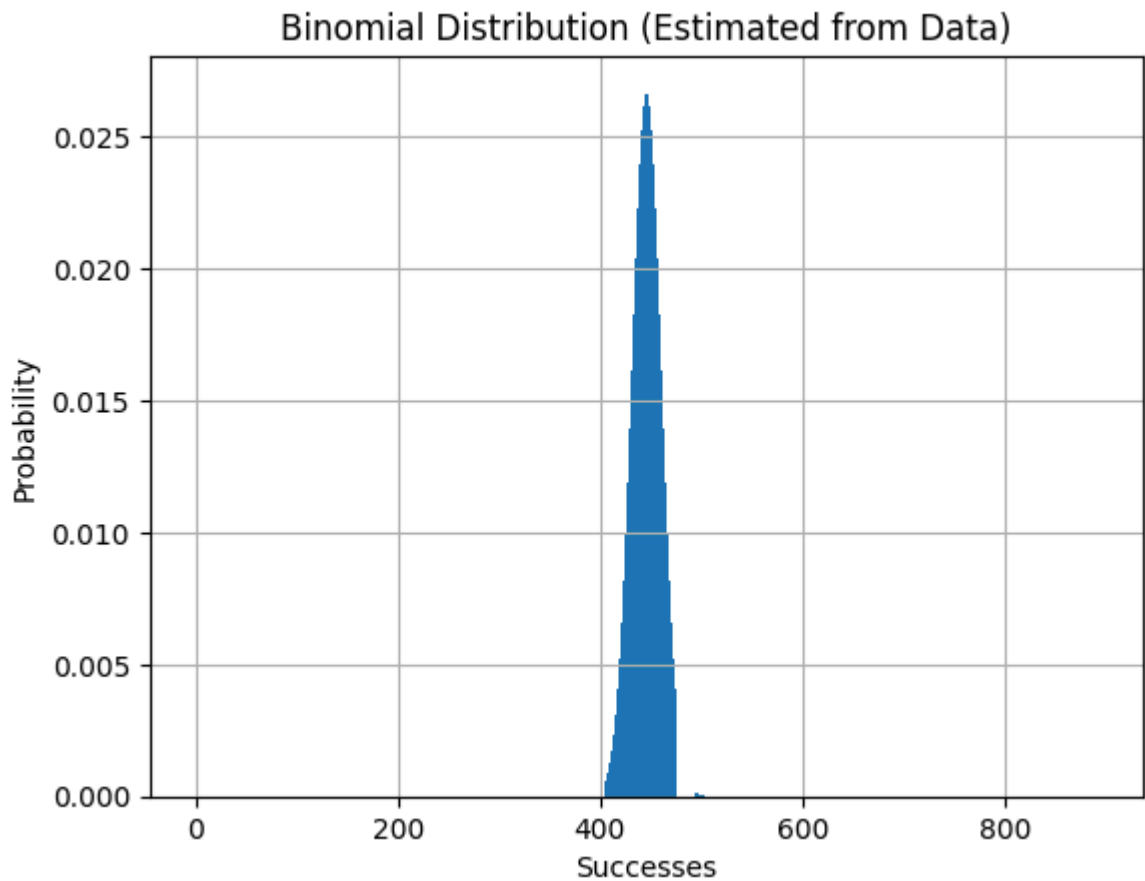
```
Selected column: PassengerId
```

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom

n = int(max(data))
p = np.mean(data) / n

x = np.arange(0, n+1)
pmf = binom.pmf(x, n, p)

plt.bar(x, pmf)
plt.title("Binomial Distribution (Estimated from Data)")
plt.xlabel("Successes")
plt.ylabel("Probability")
plt.grid()
plt.show()
```

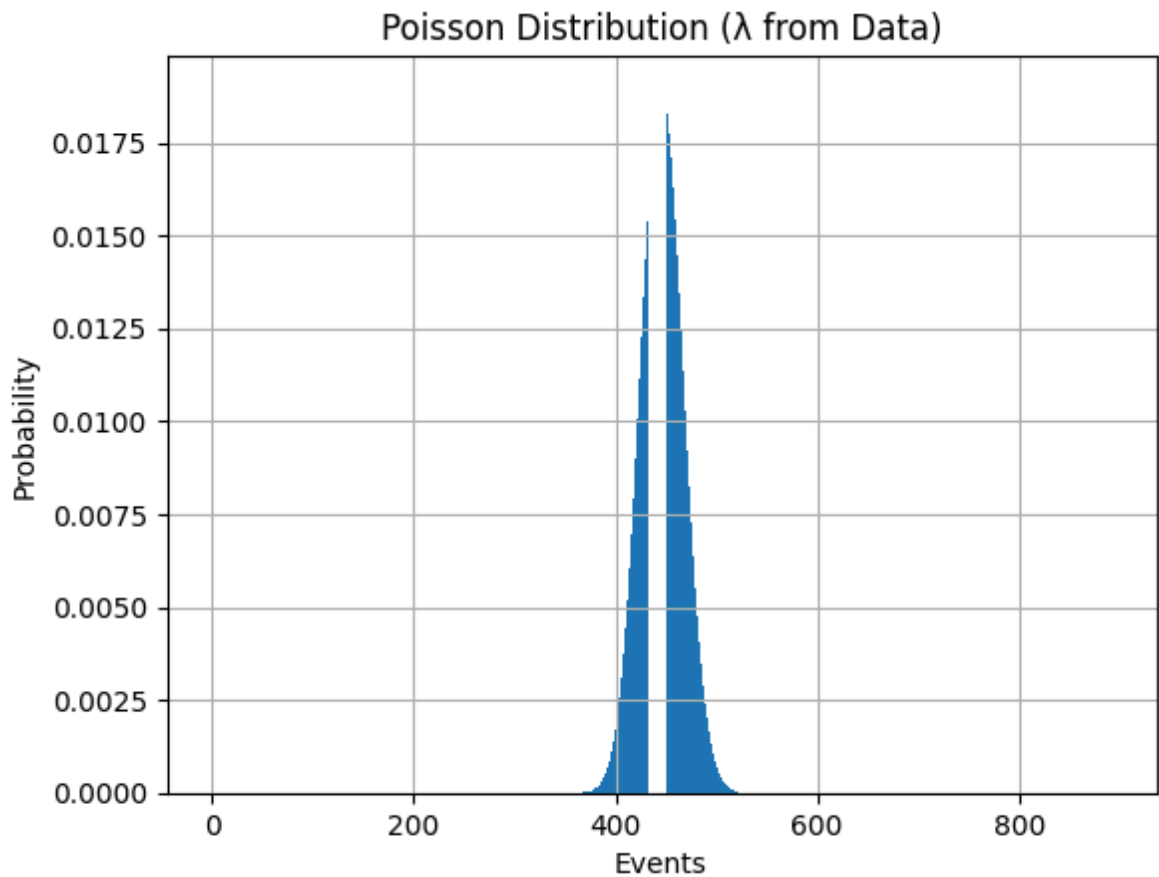


```
from scipy.stats import poisson

lam = np.mean(data)

x = np.arange(0, int(max(data))+1)
pmf = poisson.pmf(x, lam)

plt.bar(x, pmf)
plt.title("Poisson Distribution ( $\lambda$  from Data)")
plt.xlabel("Events")
plt.ylabel("Probability")
plt.grid()
plt.show()
```



```

variance = np.var(data)

print("Mean:", lam)
print("Variance:", variance)

if abs(lam - variance) < 1:
    print(" Data may follow Poisson distribution")
else:
    print(" Data does NOT follow Poisson properly")

```

```

Mean: 446.0
Variance: 66156.66666666667
Data does NOT follow Poisson properly

```

```

from scipy.stats import norm

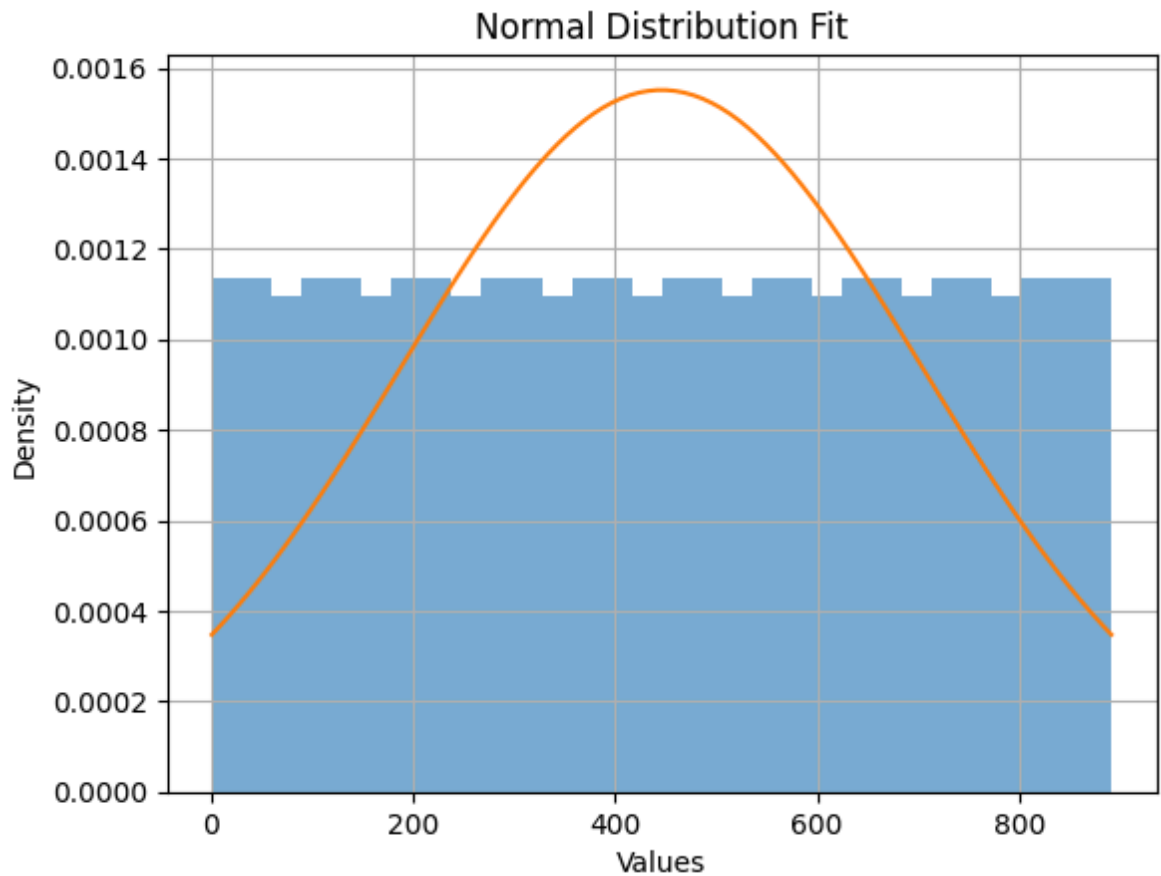
mu = np.mean(data)
sigma = np.std(data)

x = np.linspace(min(data), max(data), 100)
pdf = norm.pdf(x, mu, sigma)

plt.hist(data, bins=30, density=True, alpha=0.6)
plt.plot(x, pdf)
plt.title("Normal Distribution Fit")
plt.xlabel("Values")
plt.ylabel("Density")

```

```
plt.grid()  
plt.show()
```



```
from scipy.stats import shapiro  
  
stat, p = shapiro(data)  
  
print("p-value:", p)  
  
if p > 0.05:  
    print(" Data is approximately Normal")  
else:  
    print(" Data is NOT Normal")
```

```
p-value: 6.308055732378686e-16  
Data is NOT Normal
```

```
from scipy.stats import skew  
  
sk = skew(data)  
print("Skewness:", sk)  
  
if sk > 0:  
    print("Right skewed")  
elif sk < 0:  
    print("Left skewed")  
else:  
    print("Symmetric")
```

```
Skewness: 0.0  
Symmetric
```